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RESEARCH ARTICLE

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Restraining the formation of filter bubbles with algorithmic affordances: Toward more balanced information consumption and decreased attitude extremity

Tingting Jiang[1,2] | **Zhumo Sun**[1] | **Shiting Fu**[3]

[1]School of Information Management, Wuhan University, Wuhan, China

[2]Research Institute for Data Intelligence, Wuhan University, Wuhan, China

[3]School of Information Management, Central China Normal University, Wuhan, China

Correspondence
Tingting Jiang, School of Information Management, Wuhan University, Wuhan, China.
Email: tij@whu.edu.cn

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Abstract

In combating filter bubbles, an undesirable consequence of personalized recommendations, prior research has focused on improving algorithms to increase the diversity of the content recommended. Following a user-centered approach firmly grounded in information science, this study is dedicated to optimizing interaction patterns with algorithmic affordances, aiming to augment the diversity of the content consumed and induce favorable attitude changes. A controlled experiment was conducted on a mock personalized recommender system that provided both information and interactivity affordances, exemplified by stance labels and stance-based filters, respectively. A total of 142 participants were recruited to browse recommendations generated by the system on a specific controversial topic, and the selectivity of their information consumption behavior and the change in their attitude extremity were measured. It was found that both types of affordances were effective in reducing users' behavioral selectivity. While stance labels inhibited the consumption of pro-attitudinal information, stance-based filters facilitated the consumption of counter-attitudinal information. Furthermore, the affordances could immediately mitigate the attitude extremity of those with a higher level of algorithmic literacy. The findings not only enrich the growing body of literature on filter bubbles but also offer valuable implications for the affordance design practices of personalized recommender systems.

1 | INTRODUCTION

Personalized recommender systems, with an aim to cope with information overload, make data-driven predictions about individual users' preferences from their past behaviors and recommend relevant items they may be interested in (Zhang et al., 2021; Zhou et al., 2012). These systems are built upon various machine learning algorithms to filter and prioritize content, providing users with recommendations of news, products, songs, movies, and even social contacts at a high accuracy (del Carmen Rodríguez-Hernández & Ilarri, 2021). However, there is a growing concern that personalization reduces the diversity of information presented to users, thus creating filter bubbles where individuals are isolated from diverse ideas, beliefs, and perspectives (Pariser, 2011).

It is believed that filter bubbles exacerbate selective exposure in the current media environment and cause people to become more entrenched in their existing attitudes (Cardenal et al., 2019; Chitra & Musco, 2020). Individuals in filter bubbles may be at a higher risk of believing false or inaccurate information as they have no alternative information sources (Sun et al., 2022). Even worse, filter bubbles can lead to increasingly extreme

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attitudes and radical views of the world, contributing to polarization and radicalization (Iannelli et al., 2022). Facebook has faced criticism for its role in engendering filter bubbles and spreading misinformation through its news feed algorithm (Schwartz & Mahnke, 2021). During the 2016 U.S. presidential election, filter bubbles on social media were blamed for intensifying the ideological conflicts between supporters of the two parties (Groshek & Koc-Michalska, 2017).

In combating filter bubbles, computer scientists have focused on improving algorithms beyond traditional accuracy metrics, aiming to enhance recommendation diversity through novelty- or serendipity-based metrics (Lunardi et al., 2020; Resnick et al., 2013). Nevertheless, critiques have emerged against understanding filter bubbles purely as a technological challenge. Dahlgren (2021) emphasized that the formation of filter bubbles fundamentally involves a self-reinforcing spiral between information provision and information consumption. Algorithms respond to signals released by users' behaviors, such as clicking on a link, which may signal a preference for the linked content, by pushing more similar content to the users to match their preferences. This process can evolve into closed information universes filled with homogeneous content that caters to users' preferences while constraining their future behaviors. Therefore, the concept of filter bubbles reflects the potential negative consequences of human–algorithm interactions, rather than being solely a biased algorithmic outcome (Areeb et al., 2023; Cho et al., 2020).

Recent years have witnessed the increasing attention drawn to intervening users' interaction with recommendation algorithms by providing diversity-sensitive design (Helberger et al., 2018). Algorithmic affordances are essential design elements for incorporating diversity (Smits et al., 2024). They characterize a spectrum of explicit and implicit action possibilities that enable users to interact with, and eventually influence, the algorithms (Hekman et al., 2022). Information affordances and interactivity affordances are the two prevailing approaches to providing algorithmic affordances. It has been claimed that algorithmic affordances can promote users' awareness of information filtering and give them some sense of control over the filtering process (Bozdag & Hoven, 2015). However, there still lacks empirical evidence demonstrating the effectiveness of various algorithmic affordances for these purposes.

It should be noticed that the offering of useful algorithmic affordances is not necessarily followed by effective utilization, which may be influenced by individual differences. Algorithmic literacy (AL) has been recognized as a possible individual factor with an impact on users' interaction with algorithmically curated online environments (Dogruel et al., 2022; Oeldorf-Hirsch & Neubaum, 2025). It has been hypothesized that AL may contribute to the elimination of filter bubbles for enabling individuals to reflect on and make use of algorithm-recommended information more critically and autonomously (Dogruel, 2021; Zarouali et al., 2021). Due to limited research into AL, however, its role in moderating users' adoption of algorithmic affordances in personalized recommender systems is yet to be determined.

With an aim to reveal the potential of algorithmic affordances to debias human–algorithm interaction and hinder the formation of filter bubbles, this study conducted a controlled experiment on a mock personalized recommender system that provided both information and interactivity affordances, exemplified by stance labels and stance-based filters, respectively. The focus was determining the effects of algorithmic affordances on the selectivity of users' information consumption behavior and the change in their attitude extremity. The moderating effect of their AL was further examined. The significance of this study consists not only in its potential to enrich the understanding of the human–algorithm interaction patterns underlying the formation of filter bubbles but also in identifying opportunities for design and educational interventions to help users counter filter bubbles.

2 | LITERATURE REVIEW

2.1 | Observing filter bubbles

Due to the absence of a widely agreed definition, filter bubbles have been empirically observed using various approaches. Early research often assessed filter bubbles by examining the inter-user diversity of information provided by the same search engine, news platform, or social media. Specifically, the less overlap in content received by different users, the higher the degree to which they are segregated by filter bubbles (Bechmann & Nielbo, 2018; Dillahunt et al., 2015; Matter et al., 2022; Min et al., 2019). However, given that the filter bubble is an "individual outcome" characterized by the limited, tailored selection of information a user sees (Geschke et al., 2019), researchers have also adopted the intra-user diversity of information shown to individual users to determine if they are caught in filter bubbles (Haim et al., 2017; Liu et al., 2021).

Despite the increased diversity in information provision, users may still selectively consume information based on their existing understanding, beliefs, and values (Jones et al., 2011). People have a natural inclination to approach information that aligns with their attitudes,

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while avoiding information that contradicts them, either intentionally or unintentionally (Kim et al., 2017). Consequently, another stream of research has concentrated on the content diversity of news, music, or movies that individual users consume, indicating the presence of filter bubbles with low diversity or high selectivity in information consumption (Allen et al., 2016; Cardenal et al., 2019; Nguyen et al., 2014). A few recent studies have employed intra-user diversity of information consumed by individual users to evaluate the effectiveness of countermeasures developed to break filter bubbles (Anwar et al., 2024; Kandula, 2023).

Selective information consumption behavior is common on stance-diversified social media platforms where individuals tend to consume stance-consistent information while ignoring stance-inconsistent information (Ali & Eriyanto, 2021; Elmore et al., 2023). Gao et al. (2018) proposed the *Selective Exposure Index* (*SEI*) to indicate the selectivity of an individual's news consumption and computed *SEI* by dividing the number of stance-consistent news articles that one selected to read ($N_{\text{consistent}}$) by the total number of consistent and inconsistent news articles selected and read ($N_{\text{consistent}} + N_{\text{inconsistent}}$). This study introduces the *SEI* as a key metric for characterizing human–algorithm interaction. A higher *SEI* for an individual user suggests a greater likelihood that the interaction will lead to a filter bubble.

Some recent evidence has also suggested the critical role of individuals' own attitudes in the formation of filter bubbles. Although it is commonly assumed that Google search engine created filter bubbles by personalizing search results for individual users, the fact is Google users tended to select search queries corresponding to their own political learnings and these queries yielded own-side search results (Ekström et al., 2023). Moreover, users with more extreme political preferences exhibited higher click-through rates on political news recommendations compared to those with less extreme preferences (Liu et al., 2021). Given the complex interplay between humans' behavior and attitude (Bras & Miranda, 2013), this study argues that changes in attitude related to information consumption behavior should also be considered when determining whether a filter bubble will emerge.

Individuals with extreme attitudes may feel more comfortable in filter bubbles that limits the encountering of information challenging their preexisting attitudes in order to avoid cognitive dissonance (Knobloch-Westerwick et al., 2020). However, this will result in the reinforcement of those extreme attitudes, which, in turn, causes greater selectivity in information consumption (Dylko et al., 2018; Sude et al., 2019). Because of such self-reinforcing spiral, the increase in one's attitude extremity can be regarded as a demonstration of filter bubble formation (Montag, 2019; Tölle & Trier, 2023). Krosnick and Petty (1995) defined attitude extremity as the extent to which an attitude deviates from neutrality. Min et al. (2021) measured it with the absolute distance between an individual's attitude score and the mid-point of a Likert-type scale ranging from opposition to support.

2.2 | Algorithmic affordances

An affordance is the inherent characteristics of a physical or digital object that determine how it can possibly be used (Norman, 1995). The *Fogg Behavior Model* provides a framework for understanding how affordances impact user behavior. According to the model, a target behavior occurs when an individual has sufficient motivation and ability and is prompted to perform that behavior. Environmental cues or affordances can function as triggers for behavior, including *sparks* that boost motivation, *facilitators* that enhance ability, and *signals* that serve as reminders (Fogg, 2009).

Prior research has examined various cues provided or actions enabled by digital platforms as affordances. The existing findings indicate that affordances had a significant influence on behavioral outcomes, such as promoting gamified mobile app users' pro-environmental behavior (Zhou et al., 2023), enhancing live streaming commerce consumers' impulse purchase intention (Xie & Luo, 2021), and fostering positive changes in health-related behaviors among users of wearable activity trackers (Rieder et al., 2020). Users' attitudes toward technology use could also be affected by affordances, as mediated by trust, perceived usefulness, ease of use, or enjoyment (Jo & Park, 2023; Tsai & Ho, 2013).

Considering that algorithms are not directly visible to users, it is important for algorithmically curated information systems to incorporate algorithmic affordances into their interface design, so as to inform users how they can utilize the algorithms to achieve their goals (Shin & Park, 2019). The two major types of algorithmic affordances are: (1) *information affordances* that provide users with additional information about algorithmic outputs, including the output generation rationale and the confidence level; and (2) *interactivity affordances* that enable users to tailor algorithmic outputs to their personal needs or preferences, thereby allowing them to navigate through and formulate feedback to the outputs more effectively (Shao et al., 2020; Shin & Park, 2019). There have existed some enlightening research efforts that explored the potential of stance labels as typical information affordances and that of stance-based filters as typical interactivity affordances.

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Stance labels, a type of feature labels, are indicators of the opinions conveyed in the recommended information, promoting awareness of information with different standpoints (Gao et al., 2018). They are light-weight interface elements with the capability of nudging users to realize the existence of filter bubbles centering around their own stances (Gao, 2021). It has been found that stance labels increased the exposure to attitude-challenging information on various social topics and thus led to the acquisition of more knowledge from the opposite side of one's original stances (Liao & Fu, 2014). However, there also exists contradictory evidence indicating the undesirable effects of stance labels. Users are more likely to read pro-attitudinal news articles only while ignoring counter-attitudinal ones, because stance labels help them identify and locate the former more quickly. Even worse, labeling interventions could make fake news look more trustworthy, lowering users' perception of its extremeness (Gao et al., 2018).

It has been recognized that users should be given control over the recommendation criteria and allowed to refine the recommended items according to their preferences (Cremonesi et al., 2017). Various filters, e.g., based on topic, format, and region, are available at present to facilitate the access to and interaction with the recommendations of special interest to users (Balush et al., 2021; Hu et al., 2017; Tschinkel, 2020). Gao et al. (2017) demonstrated a pioneer attempt in creating a stance-based filter visualization for Reddit posts about the U.S. presidential election. Two different stances, i.e., supporting Donald Trump/Hillary Clinton, were represented with different colors; and clicking on the colored bubbles would result in the selection of posts with the corresponding stances. This visual filter was found useful in encouraging users to proactively explore opinions of opposite stances, which could help reduce selective exposure and attitude polarization on controversial topics.

2.3 | Algorithmic literacy

How individuals interact with various information environments is shaped not only by the available affordances but also by their own understanding of and abilities to utilize these affordances. Individual differences in information literacy, the skill set encompassing locating, evaluating, and using information, have been widely considered when predicting users' information behavior (Yu et al., 2017). With the prevalence of digital technologies, individuals' digital literacy has been playing an increasingly important role in their interaction with digital platforms and content (Noh, 2017). The evolving nature of the technological landscape drives the recent shift of attention to artificial intelligence literacy, i.e., the knowledge and skills necessary for making informed decisions about the use and impact of AI technologies and applications, as well as its fundamental subset, i.e., AL (Ng et al., 2021).

AL is a term coined to capture all the major facets that influence how users understand, use, evaluate, cope with, or even influence algorithms (Dogruel, 2021). The scale for measuring AL as developed in Dogruel et al. (2022) focuses on two dimensions: (1) *awareness of algorithm use*—the extent to which users know what algorithms can be used for and where they are actually used; and (2) *knowledge about algorithms*—the extent to which users know what types of input data algorithms take and how they select, prioritize, and present personalized content to users as outputs. AL has been regarded as a crucial factor in protecting individuals from the risks that come with algorithms, including filter bubbles (Milan & Agosti, 2019). It is promising to implement AL as a moderator in human–algorithm interaction research (Dogruel et al., 2022; Oeldorf-Hirsch & Neubaum, 2025).

The first dimension mentioned above, algorithmic awareness, has been investigated independently. According to previous findings, users with higher algorithmic awareness showed greater understanding of personalization and made more frequent adjustments to recommendation settings (Brodsky et al., 2020); and algorithmic awareness had positive effects on users' self-efficacy that further promoted their evaluation of privacy concerns, trust in algorithms, and self-disclosure (Shin et al., 2022). It has also been posited that algorithmically aware users would demonstrate higher awareness of filter bubbles and be motivated to pursue systematic processing of recommendations (Kothur & Pandey, 2023; Zarouali et al., 2021). Some initial attention has been paid to AL as a whole. It has been revealed that increasing individuals' basic knowledge, critical awareness, and abilities to use algorithms could help them cope with algorithmic curation and enable them to utilize algorithms proactively to satisfy their personal needs (Liu et al., 2023; Lv et al., 2022).

2.4 | Research questions

As reviewed above, algorithmic affordances have emerged as a promising research avenue for addressing the issue of filter bubbles, though still in its early stages.

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Feature labels and refining filters, representing information and interactivity affordances, respectively, have attracted some preliminary exploration. However, there still lacks solid conclusions regarding the optimal design and implementation of these algorithmic affordances for different user groups, especially those with different levels of AL. A major challenge that researchers have been confronted with is the absence of an established method of observing filter bubbles, which makes it difficult to assess the effectiveness of algorithmic affordances. This study contends that filter bubbles are not merely a consequence of algorithmic curation but arise from the interaction between users and algorithms that involves a self-reinforcing spiral between information consumption and attitude changes. Therefore, it is more appropriate to observe filter bubbles in terms of users' behavioral selectivity (BS) and attitude extremity.

In a stance-diversified personalized recommender system, algorithmic affordances are intended to encourage the balanced consumption of recommendations with both consistent and inconsistent stances. Such balance can contribute to the mitigation of attitude extremity because extreme attitudes often develop from the selective consumption of pro-attitudinal information. With a focus on stance labels and stance-based filters, this study examines the values of algorithmic affordances in hindering the formation of filter bubble in terms of their effects on individuals' behavior and attitudes. The following research questions are addressed:

1. How do algorithmic affordances provided by personalized recommender systems affect the selectivity of users' information consumption behavior and the change in their attitude extremity?
2. What are the roles played by AL in affecting users' interaction with algorithmic affordances?

3 | METHODS

Filter bubbles are often formed around controversial political, social, or public health topics that polarize people (Koutra et al., 2015). This study created a mock personalized recommender system that recommends information on genetically modified foods (GMFs) and investigated how the algorithmic affordances provided by the system affected individuals' behavior and attitudes. The topic of GMFs has aroused worldwide debate and led to polarized beliefs in both Eastern and Western countries. This study adopted a between-subjects design with a 2*2 factorial structure, considering the stance label as an information affordance and the stance-based filter as an interactivity affordance.

3.1 | Participants

The prospective participants for the experiment were recruited through an advertisement shared on Chinese social media platforms, targeting individuals with non-neutral GMF attitudes and selective GMF-related information consumption habits. These individuals are prone to becoming trapped in filter bubbles. Their prior attitudes (A_{prior}) toward GMFs were measured with the widely adopted GMF attitude scale (Bredahl, 2001; Appendix A) and divided into negative attitudes ($A_{\text{prior}} \leq 3$), neutral attitudes ($3 < A_{\text{prior}} < 5$), and positive attitudes ($A_{\text{prior}} \geq 5$) (Lu, 2016). They were also instructed to estimate the BS of their everyday GMF-related information consumption based on the above-mentioned *SEI* (Gao et al., 2018). Balanced consumption ($40\% < BS < 60\%$) was distinguished from selective consumption ($BS \leq 40\%$ or $BS \geq 60\%$) (LaCour, 2015).

An a priori power analysis was conducted with G*Power 3.1.9.6, indicating that a sample size of 128 would be required to achieve 80% power for detecting a medium effect size (Cohen's $f = 0.25$), at a significance level of $\alpha = 0.05$. This study successfully recruited 142 participants who met the above criteria. There were 77 male and 65 female participants, with ages ranging between 18 and 45. While 83 participants (58.45%) had completed high school, 48 (41.55%) held a bachelor's degree or higher.

The sample consisted of 76 GMF proponents and 66 GMF opponents. Each participant was assigned to one of the four groups predefined with different combinations of information and interactivity affordances and labeled with LaFa, LpFa, LaFp, and LpFp ("L"—stance labels, "F"—stance-based filters, "a"—absent, and "p"—present), respectively. The final distribution of GMF proponents and opponents among the four groups is found in Table 1. Regardless of group assignment, the participants were instructed to use the mock personalized recommender system to browse GMF-related information.

3.2 | The mock personalized recommender system

3.2.1 | Creation of messages for recommendation

To prepare the information items for recommendation, this study collected 60 real-world GMF-related messages from such popular Chinese media platforms as Pengpai[1] and Toutiao.[2] In total, 28 high-quality messages with clearly stated stances regarding GMFs were selected, including 12 pro-GMF, 12 anti-GMF, and 4 neutral

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TABLE 1 Grouping of participants.

	LaFa		LpFa		LaFp		LpFp	
Prior GMF attitude	**Positive**	**Negative**	**Positive**	**Negative**	**Positive**	**Negative**	**Positive**	**Negative**
No. of participants	19	18	20	17	19	15	18	16
Group size	37		37		34		34	

Abbreviation: GMF, genetically modified food.

messages. A professional writer was invited to rephrase the messages and their titles using simple, natural, and declarative style of language. Message lengths (300–350 Chinese characters) and title lengths (10–15 Chinese characters and excluding Arabic numbers or punctuation marks) were controlled, respectively. In addition, the publication of all messages was set to be attributed to a fictional source in the year 2023. These steps aimed to minimize the potential influences of the messages' semiotic or rhetoric characteristics. Ten master's students majoring in news and communication were then recruited to evaluate the effectiveness of the messages.

The evaluation involved reading all the 28 messages in a random sequence and completing a scoring sheet after reading each message. The scoring sheet contains five 7-point Likert-type questions that guided the evaluators to assess a message in terms of title attractiveness, message readability, logicality, and credibility, and stance about GMFs. In particular, message quality was indicated with the average of the readability, logicality, and credibility scores (Yang et al., 2019). The predefined stances of the 28 messages were consistent with the evaluators' judgment, as suggested by their average stance scores (anti-GMF ≤ 3, $3 <$ neutral < 5, and pro-GMF ≥ 5). Independent sample t tests reveal that messages with different stances were not significantly different in title attractiveness ($t_{(22)} = 0.206$, $p = 0.839$; $t_{(14)} = 1.560$, $p = 0.141$; $t_{(14)} = 0.946$, $p = 0.360$) or message quality ($t_{(22)} = -0.067$, $p = 0.947$; $t_{(14)} = -1.368$, $p = 0.193$; $t_{(14)} = -1.155$, $p = 0.267$).

3.2.2 | Design of algorithmic affordances

A mock personalized recommender system was created as a mobile application to recommend GMF-related messages. When accessing the application, users are presented with a navigation page featuring a list of hyperlinked titles. Each title is accompanied by the fictional source and a publication date. Clicking on a title would redirect users to a content page where the corresponding message is displayed. The mock system showcases a simple yet widely adopted interface design that is familiar to users.

Each participant would be provided with a personalized collection of 20 GMF-related information items as recommendations based on his or her prior GMF attitude. The collection consisted of 12 pro-attitudinal items (60%), four counter-attitudinal items (20%), and four neutral items (20%). That is, each GMF proponent/opponent would be exposed to 12 pro-/anti-GMF messages, four anti-/pro-GMF messages, and four neutral messages. The majority of the messages aligned with the participant's existing attitude, which demonstrates the personalized nature of the recommendations (Beam, 2014; Heitz et al., 2022). The inclusion of counter-attitudinal and neural messages helped increase the diversity of stances. On the navigation page, the titles in the list were arranged in a random sequence regardless of stance.

Stance labels

The information affordances of the mock personalized recommender system mainly took the form of stance labels that were placed in front of the titles on the navigation page. Given the pragmatic or argumentative function of emojis (Schneebeli, 2017), the system used a green happy/red sad/gray emotionless face to indicate the positive/negative/neutral stance about GMFs (Figure 1). The three types of stance labels could be unambiguously distinguished from one another through the application of both shape and color cues. Such stance labels capitalized on the high bandwidth of humans' visual systems and afforded effortless evaluation of the associated stances when users were quickly scanning the list of titles.

Stance-based filters

The mock system introduced stance-based filters to provide interactivity affordances. Although users were presented with "all" the 20 titles on the navigation page by default, as in Figure 1a, they were allowed to click on the "positive" (Figure 1b), "neutral" (Figure 1c), or "negative" tab (Figure 1d) to see different screens that were dynamically updated to reflect the chosen stance. Similar to the commonly seen recentness- or popularity-based filters, these stance-based filters enabled users to navigate the recommended content based on the criterion of stance. The enhanced interactivity not only empowered users with some control of the recommendation process but also

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FIGURE 1 Design of algorithmic affordances (all text translated from Chinese into English). (a) Navigation page shown to the LpFp group. (b) "Positive" tab selected. (b) "Neutral" tab selected. (b) "Negative" tab selected. F, stance-based filters; L, stance labels; p, present.

increased the transparency of the factors influencing the recommendations (Cremonesi et al., 2017).

3.3 | Measures

3.3.1 | Selectivity of information consumption behavior

This study adopted Gao et al.'s (2018) *SEI* to measure how selective the participants were in consuming the GMF-related messages on the mock personalized recommender system. Log files were generated on the mobile application's server to record users' interaction with the system, which enabled more accurate observations of their information consumption behavior, i.e., clicking on the hyperlinked titles to read the corresponding messages. For each participant, the frequencies of his or her clicks on pro-attitudinal (N_{pro}), counter-attitudinal (N_{counter}), and neutral titles (N_{neu}) were captured, respectively. Hence, the selectivity of a participant's information consumption behavior, i.e., BS, was determined by calculating the proportion of clicks on pro-attitudinal titles out of all clicks:

$$\text{BS} = \frac{N_{\text{pro}}}{N_{\text{pro}} + N_{\text{counter}} + N_{\text{neu}}}.$$

3.3.2 | Change in attitude extremity

Each participant of the experiment possessed a prior attitude (A_{prior}) toward GMFs, and his or her post attitude (A_{post}) was measured with the GMF attitude scale (Cronbach's $\alpha = 0.960$) again after reading the GMF-related messages in the recommender system. The GMF attitude score is calculated by averaging the scores of the three items in the scale. Following Min et al. (2021), this study calculated a participant's prior attitude extremity ($|A_{\text{prior}} - 4|$) and post attitude extremity ($|A_{\text{post}} - 4|$), respectively, as the mid-point of seven-point Likert-type scale is 4. The change in his or her attitude extremity ($\text{AE}_{\text{change}}$) as a result of accessing the recommender system was calculated in terms of the difference between his or her post and prior attitude extremity:

$$\text{AE}_{\text{change}} = |\text{A}_{\text{post}} - 4| - |\text{A}_{\text{prior}} - 4|.$$

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3.3.3 | Algorithmic literacy

This study used the scale developed by Dogruel et al. (2022) (Appendix B; translated into Chinese) to measure participants' AL, aligning with Kappeler (2024) and Petrovčič et al. (2024). The scale (Cronbach's $\alpha = 0.775$) comprises 11 items assessing people's awareness of algorithm use and 11 items their knowledge about algorithms. The items were designed as true or false statements. Each participant was awarded one point for each statement correctly evaluated, and his or her AL score was calculated by summing up the points obtained across all the items.

3.4 | Task and procedures

Participants' prior GMF attitudes were measured at the time of recruitment. Upon arrival at the laboratory, they were offered a brief introduction to the purpose of the mock personalized recommender system, which is to provide the public with the latest updates on GMFs, featuring the aggregation of related messages presenting different stances. Following random assignment to one of the four groups, each participant was given instructions on how to use the system. Additionally, the two groups exposed to information affordances (LpFa and LpFp) received explanations about the three types of stance labels, while the two groups exposed to interactivity affordances (LaFp and LpFp) watched demonstrations on using the stance-based filters. The training session aimed to minimize the individual differences in the experience of interacting with these interface elements, without emphasizing any particular stances on GMFs.

For the primary part of the experiment, participants were asked to perform a browsing task without a specific goal, i.e., reading GMF-related messages of interest in the system according to their daily news consumption habits, with no time constraints. Their interactions with the system were recorded with server log files. After completing the task, participants' post GMF attitudes and AL were measured with the appropriate scales. A total of 37 participants from the three experimental groups (LpFa, LaFp, and LpFp) voluntarily took part in a follow-up interview (10–20 min) to discuss their experiences with and perceptions of the algorithmic affordances provided by the system. This semi-structured interview aimed to elicit potential interpretations of the findings about the effects of algorithmic affordances, as can be seen in Appendix C.

The task and procedures were approved by the University Research Ethics Committee and strictly followed the general ethical guidelines. All participants signed a consent form and a confidentiality agreement before the experiment and received a 50 CNY/7 USD (additional 20 CNY/2.8 USD for interviewees) gift as compensation for their time upon completion.

4 | RESULTS

4.1 | Selectivity of information consumption behavior and change in attitude extremity

Thanks to the stance-diversified information environment created by the mock recommender system, approximately half of the participants (49.30%, $N = 70$) exhibited balanced information consumption behavior (i.e., $40\% < BS < 60\%$). On the navigation page, the 142 participants on average clicked on 11.43 titles (SD = 4.25, range = 3–20), including 6.51 pro-attitudinal titles (SD = 2.77, range = 1–12), 2.63 counter-attitudinal titles (SD = 1.13, range = 1–4), and 2.30 neutral titles (SD = 1.24, range = 0–4). The clicks on pro-attitudinal titles accounted for 56.96% of all clicks. This proportion is lower than that of pro-attitudinal items in the entire personalized collection, i.e., 60%, indicating the overall reduced BS.

In general, accessing the recommender system slightly increased the participants' attitude extremity (mean $AE_{change} = 0.04$). To be more specific, there were 50 participants who became less extreme in their attitudes toward GMFs, including 23 GMF proponents and 27 GMF opponents. It should be mentioned that the post attitudes of 7 proponents and 13 opponents shifted to the extent that they now fell within the range of neutral attitudes, i.e., $3 < A_{post} < 5$. However, the attitude extremity of 54 participants, including 30 proponents and 24 opponents, increased. In addition, 38 participants, including 23 proponents and 15 opponents, demonstrated no change in their attitude extremity. The overall changes in attitude extremity of all participants are found in Figure 2.

It was revealed in an independent sample t test that the GMF proponents and GMF opponents were not significantly different in BS ($t_{(137.803)} = -0.777$, $p = 0.439$) or change in attitude extremity ($t_{(118.103)} = 0.810$, $p = 0.420$). As a result, the data analysis that follows was performed on the entire sample of 142 participants without considering their preexisting GMF attitudes.

4.2 | Effects of algorithmic affordances on BS and change in attitude extremity

The two-way ANOVA was adopted to examine the main effects of the algorithmic affordances. It was revealed that

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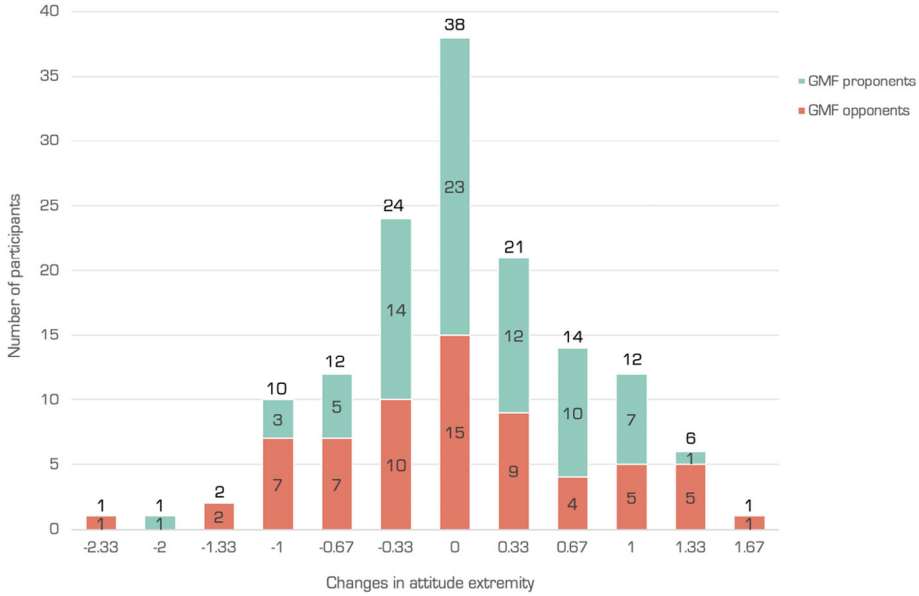

FIGURE 2 Changes in attitude extremity. GMF, genetically modified food.

both the information affordance ($F_{(1,138)} = 5.033$, $p = 0.026$) and the interactivity affordance ($F_{(1,138)} = 13.535$, $p < 0.001$) had a significant effect on the selectivity of participants' information consumption behavior. To elaborate further, the presence of stance labels ($M = 0.546$, SD $= 0.114$) led to significantly lower BS than the absence of stance labels ($M = 0.584$, SD $= 0.090$). Similarly, the presence of stance-based filters ($M = 0.533$, SD $= 0.107$) reduced BS significantly when compared to the absence of stance-based filters ($M = 0.594$, SD $= 0.093$). However, no significant interaction effect was observed between the two types of algorithmic affordances ($F_{(1,138)} = 0.871$, $p = 0.352$).

In addition, a two-way ANOVA indicted that neither the information affordance ($F_{(1,138)} = 0.998$, $p = 0.319$) nor the interactivity affordance ($F_{(1,138)} = 2.268$, $p = 0.134$) showed a significant main effect on the change in participants' attitude extremity. Also, their interaction effect was found to be insignificant ($F_{(1,138)} = 1.343$, $p = 0.248$). That is, providing stance labels and/or stance-based filters did not significantly change individuals' attitude extremity.

As mentioned in Section 3.3, BS was calculated to reflect the extent to which pro-attitudinal information was consumed relative to overall information consumption, including pro-attitudinal, counter-attitudinal, and neutral information. Given the significant main effects of algorithmic affordances on BS, this study further employed the two-way ANOVA to determine their effects on the explicitly recorded information consumption behavior, i.e., the frequencies of clicks on different titles. There existed a significant main effect of the information affordance on the frequency of clicks on pro-attitudinal titles ($F_{(1,138)} = 8.513$, $p = 0.004$). The participants clicked on significantly fewer pro-attitudinal titles when stance labels were present ($M = 5.845$, SD $= 2.208$) than when they were absent ($M = 7.169$, SD $= 3.112$). In other words, providing stance labels contributed to reducing the consumption of pro-attitudinal information. However, the frequencies of clicks on both counter-attitudinal ($F_{(1,138)} = 1.482$, $p = 0.226$) and neutral titles ($F_{(1,138)} = 0.073$, $p = 0.787$) were not significantly influenced by the information affordance.

Interestingly, a significant main effect was detected for the interactivity affordance on the frequency of clicks on counter-attitudinal titles ($F_{(1,138)} = 7.095$, $p = 0.009$). The participants clicked on significantly more counter-attitudinal titles when stance-based filters were present ($M = 2.882$, SD $= 1.015$) than when they were absent ($M = 2.392$, SD $= 1.180$). To clarify, providing stance-based filters resulted in an increase in the consumption of counter-attitudinal information. However, the interactivity affordance had no significant influence on the frequency of clicks on either pro-attitudinal ($F_{(1,138)} = 1.177$, $p = 0.280$) or neutral titles ($F_{(1,138)} = 2.578$, $p = 0.111$). Again, there was no significant interaction between the information affordance and interactivity affordance on the frequency of clicks on pro-attitudinal ($F_{(1,138)} = 0.015$, $p = 0.902$), counter-attitudinal ($F_{(1,138)} = 2.748$, $p = 0.100$), or neutral titles ($F_{(1,138)} = 0.000$, $p = 0.991$).

4.3 | Moderating effects of AL

The 142 participants had an average AL score of 15.30 (SD $= 4.06$, range $= 4$–21). Following the method outlined in Nierenberg and Dahl (2023) for categorizing high and low information literacy, this study divided the

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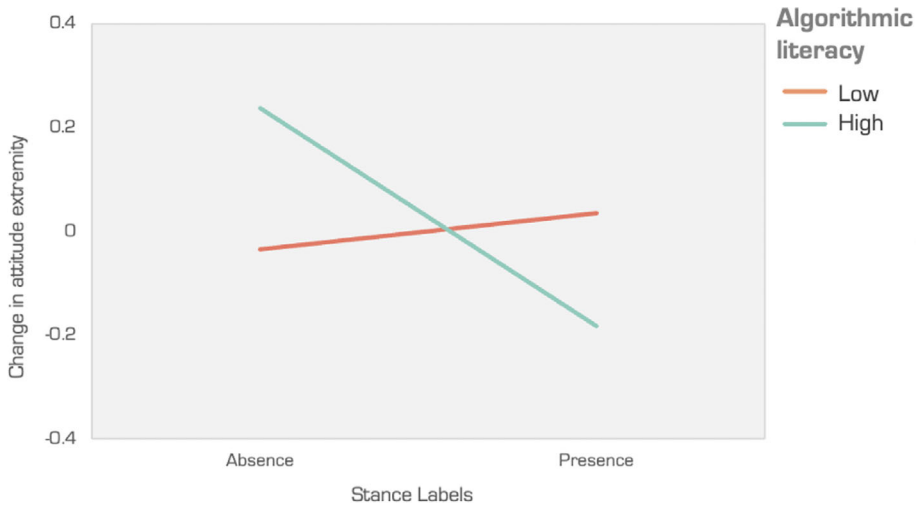

FIGURE 3 Interaction between stance labels and algorithmic literacy on change in attitude extremity.

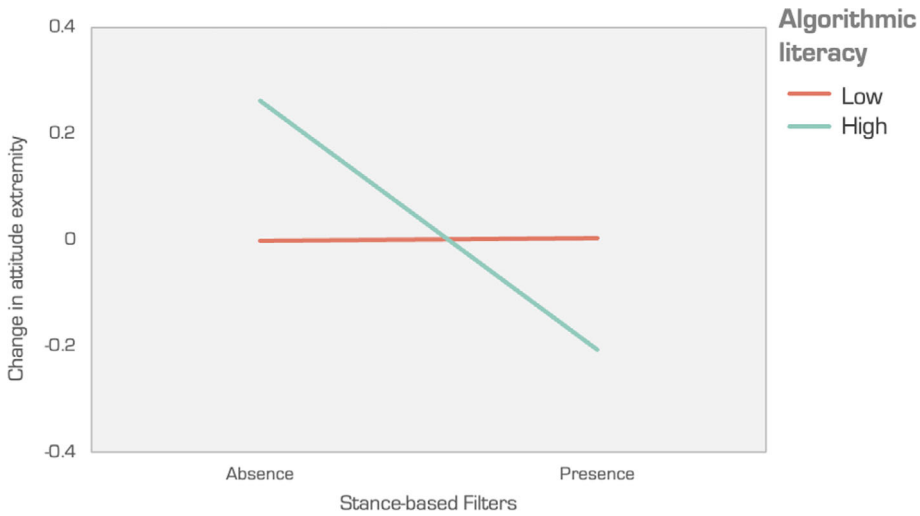

FIGURE 4 Interaction between stance-based filters and algorithmic literacy on change in attitude extremity.

sample into high-AL ($N = 67$) and low-AL ($N = 75$) groups based on a median (AL score = 16) split on the AL scores.

To assess whether AL moderates the effects of the information and interactivity affordances on BS, a three-way ANOVA was performed. The results indicate that there was no significant three-way interaction effect ($F_{(1,134)} = 0.158$, $p = 0.691$). The interactions between information affordance and AL ($F_{(1,134)} = 2.572$, $p = 0.111$) and that between interactivity affordance and AL ($F_{(1,134)} = 1.044$, $p = 0.309$) were both insignificant. As a result, the individual effects of the two types of algorithmic affordances contributed to the observed variability in BS, without AL acting as a moderator.

Again, a separate three-way ANOVA yielded no significant interaction effect of information affordance, interactivity affordance, and AL on the change in attitude extremity ($F_{(1,134)} = 1.297$, $p = 0.257$). However, the interaction between information affordance and AL was significant ($F_{(1,134)} = 4.667$, $p = 0.033$). Post hoc pairwise comparisons using the Least Significant Difference reveal the moderating effect of AL. As seen in Figure 3, providing stance labels was only effective in reducing high-AL individuals' attitude extremity ($M_{\text{absence}} = 0.237$, $M_{\text{presence}} = -0.182$, $F_{(1,134)} = 6.435$, $p = 0.012$). However, low-AL individuals' attitude extremity was not significantly changed with the presence of stance labels ($M_{\text{absence}} = -0.034$, $M_{\text{presence}} = 0.036$, $F_{(1,134)} = 0.205$, $p = 0.651$).

In addition, the interaction between interactivity affordance and AL was significant on the change in attitude extremity ($F_{(1,134)} = 4.406$, $p = 0.038$). As indicated by pairwise comparisons, AL was a moderator between interactivity affordance and change in attitude extremity. According to Figure 4, high-AL individuals' attitude extremity was significantly lower in the presence of stance-based filters ($M_{\text{absence}} = 0.262$, $M_{\text{presence}} = -0.207$, $F_{(1,134)} = 8.064$, $p = 0.005$). Again, providing stance-based filters failed to influence low-AL individuals' attitude extremity significantly ($M_{\text{absence}} = -0.002$, $M_{\text{presence}} = 0.004$, $F_{(1,134)} = 0.002$, $p = 0.968$).

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5 | DISCUSSION

5.1 | The usefulness of algorithmic affordances

In this study, RQ 1 aims to explore the influence of algorithmic affordances provided by personalized recommender systems on the selectivity of users' information consumption behavior and the changes in their attitude extremity. It was found that both stance labels and stance-based filters were effective in lowering the selectivity of individuals' information consumption behavior, indicating the usefulness of algorithmic affordances in promoting more balanced interactions, and thereby limiting the formation of filter bubbles. Affordance theory focuses on the relationship between individuals and their environment, emphasizing the functional properties of the environment and how they afford particular interactions (Zhao et al., 2021). In the present study, algorithmic affordances are functional properties of personalized recommender systems and afforded more balanced information consumption.

The effects of algorithmic affordances on user behavior observed in this study align with the *Fogg Behavior Model* as mentioned in Section 2.2 (Fogg, 2009). Stance labels visualize the stance on GMFs associated with each title. They act as *spark* triggers by capturing users' attention and arousing their interest to click on the title for more information. Differently, stance-based filters serve as *facilitator* triggers by grouping titles associated with the same stance on GMFs. They simplify information exploration through providing necessary resources or guidance. Designed to highlight the diversity of GMF-related stances, these triggers enhanced users' motivation and ability to consume information in a more balanced manner.

It is unexpected to find that stance labels fostered balanced information consumption by reducing the consumption of pro-attitudinal information. These spark triggers should be able to enhance individuals' motivation to approach stance-consistent information given humans' inherent tendency to do so (Elmore et al., 2023). For example, Gao et al. (2018) separated news feeds with opposing stances into two columns and assigned a stance label to each column, which encouraged the reading of pro-attitudinal news. In this study, differently, each title on the navigation page was preceded by a stance label. The repetitive appearance of stance labels increased the predictability of the recommended content, thus diminishing users' interest in reading further. For example, one participant in the follow-up interview noted, "*I could pretty much guess what the smiley faces would say about GMFs*" (P5), while another remarked, "*I figured I already knew enough about the risks of GMFs*" (P26). In a certain sense, such design transformed stance labels into negative spark triggers that desensitized users to stance-consistent information.

Interestingly, stance-based filters increased the consumption of counter-attitudinal information, therefore also alleviating the selectivity of information consumption. This finding basically echoes Gao et al.'s (2017) tentative evidence about the effectiveness of a visual filter in reducing selective exposure. It has been believed that humans tend to avoid stance-inconsistent information when experiencing cognitive dissonance (Jean, 2019). However, this study observed a rise in the engagement with such information that was aggregated within a single filter. A few related responses from the interviewees include "*it couldn't hurt to check it out now that the (counter-attitudinal) information was all there*" (e.g., P17), "*I hardly ever come across (counter-attitudinal) messages like this before*" (e.g., P24), and "*I wanted to see why they say GMFs are good since I didn't really buy it*" (e.g., P36). That is, with enhanced abilities to interact with information, thanks to facilitator triggers, individuals also have a desire to acquire novel information, possibly aiming to broaden perspectives and deepen critical thinking, driven by curiosity or skepticism, or just for entertainment (Peng & Liu, 2021).

5.2 | The importance of AL

For the roles of AL in affecting users' interaction with algorithmic affordances (RQ 2), this study revealed the moderating effect of AL between algorithmic affordances and change in individuals' attitude extremity. In fact, the absence of a significant main effect of algorithmic affordances on attitude change is within anticipation, because these affordances are designed to offer clues to the perception of what can and cannot be done with them, i.e., indicating the possibilities of actions. While attitude change can be an outcome of actions, it often involves more complex internal cognitive and/or affective processes instead of being a natural reaction to environmental cues (Chen et al., 2014). Possibly due to such complexity, a slight increase in attitude extremity was observed among a small proportion of the participants ($N = 54$, 38.03%).

The interesting finding is that algorithmic affordances became a direct attitude influencer when users possessed a high level of AL. Upon seeing or utilizing the

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algorithmic affordances on the mock recommender system, high-AL participants lowered their attitude extremity toward GMFs, whereas low-AL ones remained unaffected. This could be attributed to the differences in the cognitive processing approaches the two groups of users tended to employ. The Dual Processing Theory distinguishes between systematic and heuristic processing. The former approach requires more analytical, deliberate, and effortful thinking, and the latter relies on mental shortcuts to make quick judgments or decisions (Evans & Stanovich, 2013). The higher the level of an individual's AL, the more likely he or she engages in systematic processing rather than heuristic processing because of higher motivation and higher abilities to expend more cognitive efforts and resources (Jia, 2023).

In response to Q3 in the follow-up interview (Appendix C), 10 low-AL participants perceived the stance labels and/or stance-based filters "*just like other interface elements*" (P33), failing to recognize any connection between these algorithmic affordances and their attitudes toward GMFs. They were probably driven by surface-level heuristic processing to focus on the traditional functionalities of interface elements. In contrast, 16 high-AL participants acknowledged that the labels and/or filters "*were there for highlighting GMF-related stances*" (P26), thanks to the careful evaluation enabled by systematic processing. They quickly realized that these algorithmic affordances "*may be reminding users to be rational about GMFs*" (P21), because some had "*heard about the worldwide debate on GMFs*" (P18) and others "*discovered opposing perspectives on GMFs after reading the messages in the system*" (P31). The practice of reflecting on their own knowledge and actions, which can increase individuals' self-awareness, possibly contributed to the immediate change in attitude extremity among high-AL participants.

It should be noticed that the change in attitude extremity engendered by algorithmic affordances could be temporary. According to the *Elaboration Likelihood Model*, people may take two distinct routes to engage in persuasive communication, i.e., the central route and the peripheral route. The former focuses on the strength of arguments, evidence, and logical reasoning of persuasive messages, whereas the latter relies on sensory, emotional, social, and interactive cues that are appended to message content (Kitchen et al., 2014). Obviously, algorithmic affordances, for facilitating the access to content rather than presenting any content, can only affect individuals through the peripheral route. Although individuals may change attitudes based on the peripheral cues, their attitudes could be more susceptible to change under different circumstances or when exposed to different information (Liao & Huang, 2021).

5.3 | Implications

This study has several valuable theoretical implications for future research on filter bubbles. *First*, it represents a pioneering effort to shift the focus from improving recommendation algorithms to optimizing interaction patterns in combating filter bubbles. The former merely ensures diversity in content recommended, while the latter aims to increase diversity in content consumed, a more goal-directed approach. *Second*, the experimental research reported in this paper enhances the understanding of the affordance theory in the context of algorithmically curated environments. It is promising to examine the effectiveness of various algorithmic affordances and other diversity-sensitive designs in the future. *Third*, this study introduces the concept of "attitude extremity," proposing that the observation of filter bubbles should encompass not only external behavior but also internal attitudes. This theoretical perspective emphasizes the complex formation process of filter bubbles as they evolve through a self-reinforcing spiral.

This study also presents useful practical implications for effective human–algorithm interaction. On the one hand, personalized recommender systems can consider incorporating algorithmic affordances, e.g., stance labels and stance-based filters, into interface design to foster a more balanced and varied information diet for their users. A critical ethical concern in this context involves the platforms' responsibility to offer substantial transparency and user control over their algorithms with an aim to minimize the emergence of filter bubbles. On the other hand, there is a need for educational interventions to enhance users' AL. It is important to promote users' awareness and understanding of filter bubbles and encourage them to reflect on potential biases in their own use of personalized recommender systems. Platforms can collaborate with educators to cultivate a more informed and capable user base. However, persuading platforms to adopt these diversity-enhancing strategies pose a great challenge due to the increased costs and requires further investigation.

6 | CONCLUSIONS

In tackling the issue of filter bubbles, this study embraced a user-centered approach firmly grounded in information science, with a dedicated focus on examining the usefulness of algorithmic affordances. The selectivity of users' information consumption behavior and the change in their attitude extremity, as a result of their interaction with a mock personalized recommender system, were observed in a controlled experiment. The two typical

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algorithmic affordances, i.e., stance labels and stance-based filters, were both found to be effective in eliciting more balanced information consumption. In the presence of these affordances, individuals with a high level of AL further became less extreme in their attitudes toward the given topic.

The contributions of this study are threefold. First, this study highlights the trend of understanding filter bubbles as a negative consequence of human–algorithm interaction and establishes a novel framework for observing filter bubbles in terms of users' BS and attitude extremity. Second, this study provides empirical evidence about the usefulness of algorithmic affordances as well as the importance of AL, engendering insightful implications for eliciting more balanced, informed, and equitable human–algorithm interaction. Third, this study expands the methodological horizons for future research by proposing specific methods for quantifying users' BS and attitude extremity, allowing for a more accurate characterization of interaction patterns.

Nevertheless, several limitations should be acknowledged. First, despite the usefulness of algorithmic affordances in promoting more balanced information consumption, some participants' attitude extremity even increased slightly after consuming the diverse information provided by the mock system. This indicates a need for further research to explore the self-reinforcing spiral between information consumption and attitude changes. Second, this study focused on a specific controversial topic (i.e., GMFs) and two common forms of algorithmic affordances (i.e., labels and filters), which may limit the generalizability of the findings to various real-world contexts. Future research could investigate additional topics often linked with filter bubbles and examine the effectiveness of other forms of algorithmic affordances, such as stance-related visualizations or customization controls, to engender a more comprehensive understanding of user–algorithm interaction. Last but not the least, the experimental design adopted by this study only allowed for the observation of users' information behavior and temporal attitude changes on a single occasion, though the formation and disruption of filter bubbles may both involve a more extended timeframe. Longitudinal studies should be conducted to assess the long-term impact of interventions on filter bubbles.

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ENDNOTES

[1] https://www.thepaper.cn.

[2] https://www.toutiao.com.

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APPENDIX A: GENETICALLY MODIFIED FOOD ATTITUDE SCALE

1. Applying gene technology in food production is *1 = extremely bad–7 = extremely good*.
2. Applying gene technology in food production is *1 = extremely foolish–7 = extremely wise*.
3. I am *1 = strongly against–7 = strongly for* applying gene technology in food production.

APPENDIX B: ALGORITHMIC LITERACY SCALE

B.1 | ALGORITHMIC KNOWLEDGE SCALE

What do you think algorithms do on the Internet? (True, False, Don't know)

1. Algorithms recognize that results, such as e.g., search results, are incomplete and automatically correct themselves.
2. Algorithms can develop themselves in a completely different direction from that for which they were created.

You will now see some statements about algorithms, some of them are true, some are false. Please make your judgment. (True, False, Don't know)

3. I can influence algorithms with my internet usage behavior.
4. The database used by an algorithm is not decisive in determining its quality.
5. When searching for a job online, job offers displayed may vary from person to person despite the same search entry.
6. The use of algorithms which deliver personalized content can mean that the content you find is mostly consistent with your preexisting opinions.
7. Algorithms are able to think like human beings.
8. Algorithms are independent of government censorship.
9. Algorithms present both chances and risks.
10. For some media companies, content that is repeated regularly (e.g., traffic reports) is already written by algorithms.
11. Humans are never involved when algorithms are used.

B.2 | ALGORITHMIC AWARENESS SCALE

There is a large amount of data that can be used in the development and application of algorithms. Here you can see a selection of possible sources. (Is used, Is not used, Don't know)

12. Smart speaker (e.g., Alexa)
13. Smart TV
14. Wearable computing devices such as activity trackers, heart rate monitors
15. Internet browsers (e.g., Internet Explorer, Firefox, Opera, Google Chrome)
16. Electronic payment (credit, debit cards)
17. Cell Phone Towers
18. Computer Games

Algorithms are already being used in very different areas. Do you know which of the following functions are often performed by algorithms? (Are performed, Are not performed, Don't know)

19. To create weather forecasts
20. To make product recommendations
21. To create financial news (stock markets)
22. To personalize advertisements

*Wrong answers underlined.

APPENDIX C: SEMI-STRUCTURED INTERVIEW GUIDE

1. How did you use the stance labels and/or stance-based filters when accessing the personalized recommender system?
2. Do you think that the stance labels and/or stance-based filters have influenced your consumption of the genetically modified food-related messages? Why or why not? How?
3. Do you think that the stance labels and/or stance-based filters have influenced your attitudes toward genetically modified foods? Why or why not? How?
4. What are your thoughts on the stance labels and/or stance-based filters? Do you have any suggestions for improving their design?

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